VITAL: An End-to-End Contactless Vitals-to-Triage Pipeline with Hybrid rPPG Correction and a Measured Case for Deterministic Clinical Safety

Kushal Raj G S
Dept. of AIML
BMS Institute of Technology and Management
Bengaluru, India
kushalrajgs@gmail.com

Milind K
Dept. of AIML
BMS Institute of Technology and Management
Bengaluru, India
milind_k05@yahoo.com

M Manoj
Dept. of AIML
BMS Institute of Technology and Management
Bengaluru, India
manoj044m@gmail.com

Pradeep K R
Dept. of AIML
BMS Institute of Technology and Management
Bengaluru, India
pradeepkr@bmsit.in

Anupama H S
Dept. of AIML
BMS Institute of Technology and Management
Bengaluru, India
anupamahs@bmsit.in

M Krishna Prasad
Dept. of AIML
BMS Institute of Technology and Management
Bengaluru, India
krishnamuralimk08@gmail.com

***Abstract*—Contactless heart-rate sensing from a webcam is evaluated on its own, and is increasingly paired with language models trusted to reason about clinical risk straight from raw vitals. VITAL puts both assumptions to the test. Its rPPG pipeline – CHROM corrected by a small gradient-boosted model – reaches 6.48 BPM mean absolute error across 71 real sessions from UBFC-rPPG and UBFC-Phys, down from 9.43 BPM for the classical signal alone, with most of that gain landing exactly where three separate signal-processing fixes could not reach: motion artifact from talking. Separately, we asked whether VITAL's language-model narration layer could be trusted to simply repeat a number it was handed. Across 16 runs on 8 controlled clinical scenarios, it echoed the correct priority score verbatim only 6.2% of the time, even when explicitly told not to recompute it. That result is why a deterministic scoring engine, not the language model, remains the sole source of truth for acuity, shock status, and priority in VITAL. We report what worked, what failed, and where this system still trails the closest published rPPG benchmark.**

***Index Terms*—Remote photoplethysmography, rPPG, CHROM, hybrid machine learning, emergency triage, clinical decision support, multi-agent LLM systems, deterministic safety, contactless vitals monitoring**

I. Introduction

A. Background and Motivation

Emergency departments triage patients under time pressure with a static, sensor-dependent workflow; delays in acquiring even basic vitals (heart rate, respiratory rate) for a waiting patient directly contribute to missed deterioration. **Remote photoplethysmography (rPPG)** extracts heart rate, HRV, and respiratory rate from an ordinary webcam feed, without attaching a sensor to the patient at all [2], [3], [9]. At the same time, **large language models** are increasingly proposed as the reasoning layer for turning those vitals into a triage decision, orchestrated through multi-agent frameworks assigning perception, diagnosis, and coordination roles to separate LLM agents [14], [15]. This is attractive – a webcam plus an LLM API replaces a nurse's stopwatch and a static rule table – but it raises a question rarely measured directly: when an LLM agent is told a patient's exact clinical numbers and instructed only to restate them, **does it actually do so?**

B. Problem Statement

Two gaps motivate this work. **First**, rPPG systems are frequently validated once, on a synthetic or clean benchmark signal, without the harder work of tracing why specific real subjects fail and fixing the underlying cause rather than tuning a global threshold. **Second**, and more consequentially, agentic-AI clinical tools increasingly let an LLM see raw vitals and reason toward a severity classification directly [13], without measuring whether it reliably preserves a number it was only supposed to **restate, not recompute**. Warrier and Abhilash's own agentic orchestration framework for hospital workflows [14] builds in a runtime assurance layer to veto unsafe agent actions – a guardrail need their architecture addresses procedurally, **not empirically**.

C. Contributions

This paper makes five contributions, each backed by a measured result rather than a claim:

1) A complete, deployed contactless-vitals-to-triage pipeline: webcam capture, **MediaPipe** ROI extraction, a **CHROM** rPPG engine, a **deterministic clinical scoring engine**, a three-agent **CrewAI** narration layer, and a Next.js/Flask/Neon Postgres stack – running end to end, not a benchmarking script.

2) **Three independent classical-DSP bugs** in the rPPG engine, each root-caused by tracing a specific failing subject rather than tuning a global parameter: a harmonic lock-on bug, a hardcoded-frame-rate bug, and a history-aggregation freeze bug. Combined, they took the real-video baseline from **17.4 to 8.4 BPM MAE** on the original 56-session set.
3) A **hybrid classical-DSP + gradient-boosted correction layer**, validated with subject-level **GroupKFold**, that reduces real-video MAE from **9.43 to 6.48 BPM (31% relative reduction)** across **71 sessions and 51 subjects**, with the largest gain in the one condition (speech) DSP-level fixes never touched.
4) A deterministic scoring engine that expresses **Compensated Shock** as a **multiplicative interaction** between tachycardia and tachypnea ramps rather than a boolean AND-gate, and an explicit rule – validated, not assumed – that this engine is the **sole source of truth** for the numbers that drive triage priority.
5) The first direct, repeated measurement of whether an LLM agent faithfully restates a given clinical number: across **16 real CrewAI runs**, the priority score was restated correctly only **6.2% of the time**, even under an explicit "do not recompute" instruction – turning the guardrail need identified in [14] into a concrete number.

II. Related Work

A. rPPG Signal Extraction

CHROM [2] and **POS** [3] remain the two dominant classical rPPG projection methods. Pirzada et al.'s **2024 survey of over 300 rPPG studies** documents both remain sensitive to motion artifact and ambient-light variation, with the field still overwhelmingly validated in labs rather than realistic settings [9] – the same sensitivity this project traces in its own speech-condition failures. At the other extreme, a **2026 clinical validation** of a commercial system in 47 cardiovascular patients reports **1.06 BPM MAE** against ECG [10]; the gap between that and this project's real-video numbers (Section VI) is a large part of why this paper does not overstate its own.

B. ML-Corrected and Hybrid rPPG

The closest published comparator to this project's hybrid layer is **Qaadan et al.**, who combine five rPPG methods with four classical regressors across four datasets under a single 80/20 split [1]. Their best result on the UBFC-rPPG Part 2 subset – CHROM into XGBoost with permutation-importance feature reduction – reaches **4.60 BPM MAE / 5.57 RMSE**, currently ahead of this project's comparable numbers (Section VI). Their own conclusion names **"hybrid algorithmic-learned models"** as unbuilt future work [1]; that is this project's actual architecture.

C. Agentic and Learning-Based Emergency Triage

A 2025 systematic review of interpretable ML for ED triage found **gradient-boosted models with SHAP or LIME** dominate the field (AUROC 0.78–0.96), but identified persistent gaps in **operational-metric evaluation, fairness auditing, and data diversity** beyond high-income countries [11]. Yaddaden et al. use a **Polynomial Features transform** for non-linear interactions between clinical variables, reporting **99.80% accuracy** for ESI-level prediction on MIMIC-IV [12] – the same principle this project's engine applies at smaller scale, multiplying a tachycardia ramp by a tachypnea ramp rather than OR-ing two booleans. Sharma et al.'s **Triage-Bot** is the most comparable deployed system: an rPPG extractor with speech intake feeding a rule-based CTAS classifier [15]; its evaluation is still pending recruitment. Liu and Tsai combine post-hoc explainability with a **FAISS/OpenAI-embeddings RAG pipeline** over nursing guidelines [13]; VITAL's own retrieval is deliberately simpler (term-frequency cosine similarity). None of these measures whether an LLM layer's output **reproduces a given clinical number faithfully**.

D. Research Gap

No system we reviewed combines root-caused DSP fixes, a subject-level cross-validated correction layer, an enforced deterministic/LLM separation, and a repeated measurement of narration faithfulness. The correction layer closes a capability named as unbuilt future work by the closest rPPG benchmark [1]; the faithfulness measurement turns a guardrail need the agentic-orchestration literature addresses only procedurally [14] into an actual number.

III. System Architecture

A. Overview

VITAL is organized as **three layers** against a live Flask API and Next.js dashboard, backed by Neon Postgres. A **perception layer** extracts a face ROI via MediaPipe Face Mesh (Haar-cascade fallback) and feeds normalized RGB means into an rPPG DSP engine. A **clinical-reasoning layer** computes ESI-style acuity, a Compensated Shock flag, and a priority score deterministically. A **narration layer** – a three-agent CrewAI pipeline – restates that output for the dashboard and for ARIA, a voice-enabled assistant with read-only database access. Fig. III summarizes the pipeline.

B. Perception Layer

MediaPipe's [6] **468-point Face Mesh** isolates forehead and malar-cheek landmarks with yaw-aware gating so an off-angle cheek region is excluded rather than averaged in noisily. The layer also estimates **ambient luminance** and **centroid drift** per frame, exposed to the correction model as candidate features.

C. rPPG DSP Engine

The default projection is **CHROM** [2]: per-channel normalization, projection onto two chrominance axes, and adaptive scaling by the projected-signal standard-deviation ratio. A **Welch periodogram** locates the dominant heart-rate frequency; **cubic-spline upsampling to 250 Hz** supports RMSSD-based HRV; a bandpass (0.10-0.50 Hz) extracts respiratory rate. **POS** [3] is a hot-swappable alternative, **not wired into any live route** (Section V-C).

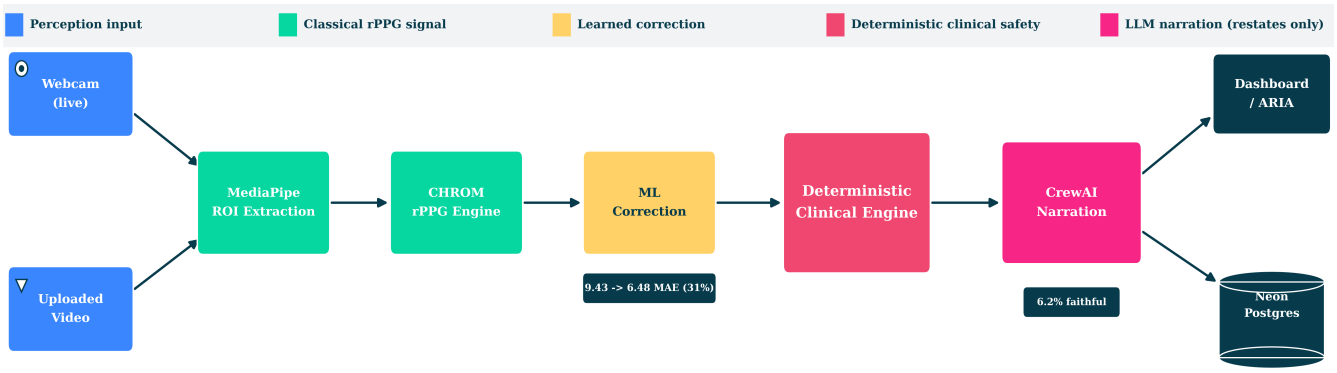


Fig. 1. VITAL's end-to-end pipeline. Blue: Perception input (webcam or uploaded video). Teal: The rPPG signal chain. Amber: the learned correction layer. Coral: the deterministic clinical engine – the sole source of truth for ESI-style acuity, the Compensated Shock flag, and priority score. Pink: the CrewAI narration layer, which only restates and explains that output and never computes or overrides it. Badges mark the two numbers that motivate this separation: a 31% MAE reduction from the correction layer, and 6.2% narration faithfulness for the one field a prior revision let the LLM compute unsupervised (Sec. V-D).

D. *Deterministic Clinical Engine*

The **deterministic clinical scoring module** computes **ESI-style acuity**, the **Compensated Shock flag**, and a **1-100 priority score** directly from extracted vitals in plain deterministic code – **not sampled from an LLM**. Rather than a boolean gate that simply checks whether heart rate exceeds 100 bpm and respiratory rate exceeds 20 breaths/min, which treats a barely-elevated reading identically to a severe crisis, severity above that gate is graded by a **multiplicative interaction**:

$$\text{tachy}_{HR} = \text{ramp}(HR,\ \text{threshold} = 100,\ \text{span} = 40) \quad (1)$$

$$\text{tachy}_{RR} = \text{ramp}(RR,\ \text{threshold} = 20,\ \text{span} = 15) \quad (2)$$

$$\text{shock}_{\text{interaction}} = \text{tachy}_{HR} \times \text{tachy}_{RR} \quad (3)$$

where $\text{ramp}(v, t, s)$ rises linearly from 0 to 1 as v climbs from t to $t + s$. Multiplying rather than averaging or OR-ing means **risk accumulates only when both vitals are elevated together**, matching the physiological definition of *compensated* shock as a combined cardiorespiratory response. HRV (RMSSD) and a **Baevsky Stress Index** are folded in as secondary, lower-weight terms, and **any Compensated Shock flag unconditionally forces mandatory manual review**. We describe this deliberately as a **custom, vitals-only scoring rule**, not a faithful ESI [7] or NEWS2 [8] implementation: both require inputs (chief complaint, SpO_2, blood pressure, temperature) unmeasurable via webcam alone.

E. *Multi-Agent Narration Layer*

Three CrewAI agents run in sequence: a **Perception Agent** restates already-extracted vitals; a **Diagnostic Agent** explains the ESI level and shock flag, grounded in retrieved guideline context; a **Coordinator Agent** restates the priority score for the queue. Guideline grounding uses **local term-frequency cosine similarity**, simpler than comparable RAG pipelines [13]. Each request builds a **fresh set of agents, tasks, and a crew**, since CrewAI's task objects store output internally and are not safe to re-enter concurrently.

Why the LLM does not compute the ESI level. An earlier revision let the agents freely generate the ESI level and shock flag as JSON, falling back to a hardcoded default of 3 on any parse failure – making **the single most safety-critical number in the system non-deterministic**. Section V-D presents the measurement that motivated reverting this. The live camera path holds one global camera/session (one workstation, one webcam) and is single-stream; uploaded-video runs execute concurrently.

F. *Dashboard Interface*

Fig. 2 shows the two primary dashboard views: **Vitals Monitor** (left), exposing the live intake feed, optical physiology indicators, and rPPG waveform panel; and **Triage Dispatch** (right), the central queue sorted by acuity. Both are shown idle (hardware disconnected, queue empty), captured before a live acquisition run.

IV. Dataset Description

Real-video validation uses **71 sessions across 51 unique subjects**, pooled from two public datasets [4], [5]:

- **UBFC-rPPG** [4] – **41 sessions** (both public releases combined), a math-stress protocol with ground truth from a contact pulse-oximeter waveform (`gtdump.xmp`/`ground_truth.txt`).
- **UBFC-Phys** [5] – **30 sessions** (10 subjects × 3 conditions: rest, speech, arithmetic, following a Trier Social Stress Test-inspired protocol), ground truth from a wrist-worn blood volume pulse (BVP) waveform, peak-detected into a per-session BPM.

Class composition across conditions is uneven by design (10 rest, 10 speech, 10 arithmetic, 41 unlabeled UBFC-rPPG), reported stratified rather than pooled into a misleading average (Table IV). All model validation uses **GroupKFold by subject identifier**: no subject's sessions appear in both folds, preventing the correction model from **memorizing a specific person** rather than learning a general correction.

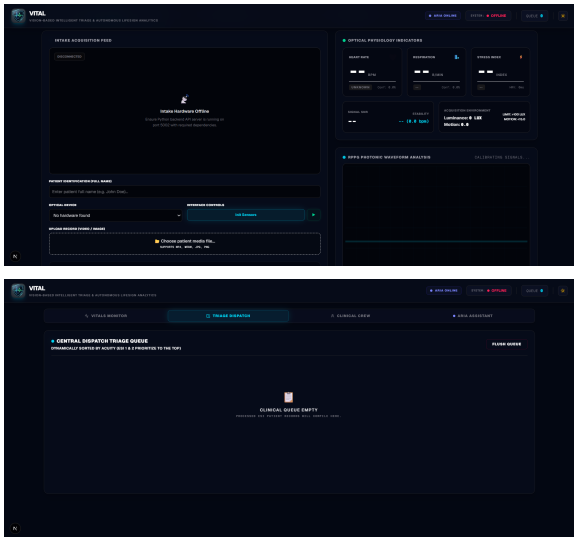


Fig. 2. VITAL's dashboard interface. Top: the Vitals Monitor view, showing the intake feed, optical physiology indicators, and rPPG waveform panel. Bottom: the Triage Dispatch view, showing the acuity-sorted central queue. Both panels are captured in their idle state (no active acquisition session), not populated with live measurement data.

V. Methodology

A. *Classical DSP Bug-Fixing, By Tracing Individual Failures*

Rather than tuning global thresholds against an aggregate metric, each bug below was found by tracing one specific failing real subject to a root cause, then verified against the full real-video set. Table I summarizes the three found bugs.

TABLE I
Classical DSP Bugs, Root-Caused and Fixed

Bug	Root Cause	Fix, Verified Effect (BPM)
Harmonic lock-on	Peak-picker locks onto half the true rate for sustained stretches	Promote to the double when it has comparable power; subj.41 52.6→8.2, subj.46 41.2→1.3, subj.9 60.9→2.8
Hardcoded frame rate	Every engine assumed 30 fps regardless of the real source rate	Read the true source fps; subj.25 23.5→3.5, subj.26 25.5→2.5
History freeze	An 18 BPM noise-rejection gate can anchor on an early wrong burst forever	Allow re-anchoring only while the anchor is still small; subj.37 21.0→6.0, no regression elsewhere

The history-freeze fix earned a second look: a first attempt fixed its target subject but **silently regressed the harmonic-lock-on fix's flagship case** back to a 41.8 BPM error, revealing the earlier "fixed" result had actually been propped up by this same freeze bug anchoring correctly by luck, not design. The corrected, narrower fix resolved both without re-breaking either. Combined, all three fixes moved the original 56-session real-video baseline from **17.4 to 8.4 BPM MAE (52% relative reduction)**; RMSE 24.1→15.6.

Fig. 3 shows this fix's effect on the five subjects it was verified against.

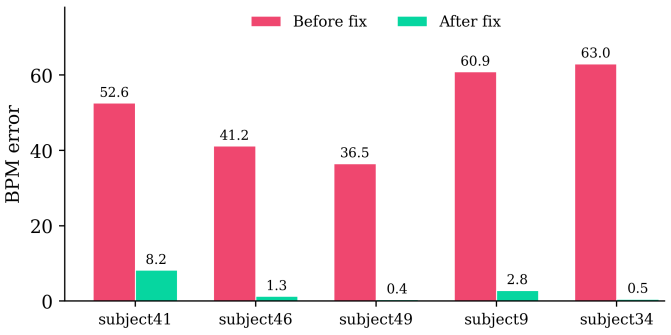


Fig. 3. Per-subject BPM error before and after the subharmonic-promotion fix for the harmonic lock-on bug. Zero effect on the shipped CHROM synthetic MAE (0.29, unchanged).

B. *Hybrid Classical-DSP + ML Correction*

A **gradient-boosted regressor** (`GradientBoostingRegressor`, **Huber loss**) is trained to correct the classical CHROM pipeline's per-cycle BPM output, using **ten features per cycle**: raw BPM, signal-quality index, SNR (dB), a stability score, respiratory rate and its confidence, HRV, the Baevsky stress index, and the adaptive filter's low/high cutoff frequencies. The model is evaluated with **5-fold GroupKFold by subject**, and every session-level result reported in Section VI uses a model that **never saw that subject during training**. Huber loss was adopted after comparing it against squared-error loss on the same folds; it produced a small but consistent improvement (**8.80→8.67 BPM** per-cycle MAE, **6.65→6.48 BPM** session-level MAE), consistent with squared-error over-weighting a handful of extreme-outlier sessions. Larger models – a bigger regularized neural network, a bigger gradient-boosted model, and an ensemble of both – **did not outperform the small deployed model**; the binding constraint is **51 unique subjects**, not the number of labeled cycles.

C. *POS's Own Bug, Partially Fixed*

POS [3] exhibits a **mirror-image failure to CHROM's**: it overshoots to roughly double the true rate from cold start, rather than undershooting. A symmetric "demote to the half" check, added to the same peak-selection logic as CHROM's fix, cut POS's synthetic MAE from **26.71 to 16.00 BPM (28.88 RMSE)** with zero effect on CHROM's own numbers – a **real, partial fix**: two of seven synthetic cases still fail. On real video (tested for the first time here), results were mixed: one subject was accurate (1.7 BPM error), but two others failed in opposite directions, and **threshold tuning could not fix either** – one subject's true rate never appears as a spectral candidate at all, and the other's correct candidate is present but weak (median **11% power**, never crossing an aggressive 30% threshold). This traces to a **structural gap**: CHROM applies global mean-normalization and per-buffer variance equalization before projecting; POS applies only per-window mean-normalization, consistent with its literature description as lacking CHROM's stronger illumination compensation [1].

Closing this needs a rewrite of POS's projection math, not its peak-selection heuristics; it remains **experimental and out of the live product**.

D. Agent Narration Faithfulness Evaluation

To test whether the LLM narration layer can be trusted with clinical numbers directly, we ran the real CrewAI pipeline (live API calls to an **8-billion-parameter** instruction-tuned LLM, not a mock) against **8 controlled clinical scenarios** spanning the acuity range – normal, isolated tachycardia, isolated tachypnea, mild shock, severe shock, bradycardia, low signal-quality, and no-signal – each run twice (**16 total runs**), checking whether the LLM's final JSON output **exactly matched the deterministic assessment on every field**, despite the task instructions explicitly stating to restate the given values verbatim and not recompute them.

VI. Experimental Results

A. Synthetic Self-Test

Table II isolates the DSP math from camera capture and face detection, using a physiologically-modeled synthetic RGB trace across seven known-BPM trials (50-140 BPM, 2% sensor noise).

TABLE II
SYNTHETIC VALIDATION (DSP CORE ONLY)

Algorithm	MAE (BPM)	RMSE (BPM)
CHROM (shipped default)	**0.29**	0.53
POS (experimental, not wired in)	16.00	28.88

B. Real-Video Classical vs. Hybrid Accuracy

Table III reports the headline result across all 71 real sessions; the correction model is always tested on subjects it has never seen before, so it can't simply have memorized their patterns.

TABLE III
REAL-VIDEO ACCURACY, 71 SESSIONS / 51 SUBJECTS

Pipeline	MAE (BPM)	RMSE (BPM)
Classical DSP only (CHROM)	9.43	16.51
Hybrid (DSP + ML correction)	**6.48**	**9.66**
Relative MAE reduction	31%	

TABLE IV
PER-CONDITION ACCURACY BREAKDOWN

Condition	Classical	Hybrid	n
Speech (talking)	21.0	**7.8**	10
Arithmetic (mental stress)	10.4	**7.0**	10
UBFC-rPPG (unlabeled task)	7.4	6.1	41
Rest	5.1	6.1 (worse)	10

The hybrid model helps every condition except already-clean rest, where the classical baseline has little bias left to correct (Fig. 4). **Its largest single gain is in speech** – the one condition

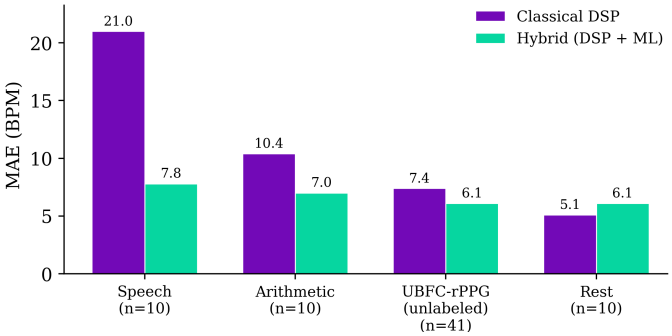


Fig. 4. Classical vs. hybrid MAE by condition. The hybrid model's gain is concentrated in speech and arithmetic; on already-clean rest it is slightly worse, since there is little systematic bias left for it to correct.

three independent DSP-level bug fixes never touched, since mouth-movement motion artifact is a fundamentally different failure mode from frequency-domain harmonic ambiguity or a frame-rate mismatch.

C. Comparison Against the Closest Published Benchmark

Table V restricts the comparison to the UBFC-rPPG-only subset directly comparable to Qaadan et al.'s evaluation [1], since their study does not include UBFC-Phys.

TABLE V
COMPARISON ON THE UBFC-RPPG-ONLY SUBSET

Method	MAE (BPM)
This work – classical DSP only	7.42
This work – hybrid (DSP + ML correction)	6.12
Qaadan et al. – CHROM + XGBoost (their best) [1]	**4.60**

We report this honestly: on the narrow task of rPPG heart-rate accuracy alone, **their approach currently beats both of ours**, and not by a rounding margin. Their own permutation-importance analysis repeatedly ranks **skewness, lag-1 autocorrelation, sample entropy, and spectral centroid** among their top CHROM features; none of these are in this project's ten-feature correction schema, which is a **concrete, named, and currently unclosed gap**. What this project has, that a pure benchmarking study does not, is the remainder of this section.

D. Agent Narration Faithfulness

Table VI reports the field-level exact-match rate between the LLM's restated output and the deterministic ground truth it was given, across all 16 runs.

Even under an explicit "restate this exact number, do not recompute it" instruction, **the LLM almost never produced a clean, correct integer** for the priority score – it repeatedly embedded descriptive text in the field instead (e.g. `"priority_score": "ESI Level 2 -- Emergent / Compensated Shock (score=0.311)"`). A related finding from the same runs: the LLM occasionally **attempted to call tools that do not exist** in this system at all, a second, independent reliability gap in the agent layer.

TABLE VI
LLM AGENT NARRATION FAITHFULNESS (16 RUNS, 8B-PARAMETER LLM)

Field	Faithfulness rate
JSON parses at all	15/16 (93.8%)
ESI level restated correctly	13/16 (81.2%)
Priority score restated correctly	**1/16 (6.2%)**
Shock flag restated correctly	15/16 (93.8%)
Diagnosis text – exact wording	6/16 (37.5%)
Diagnosis text – correct meaning, different words	12/16 (75.0%)

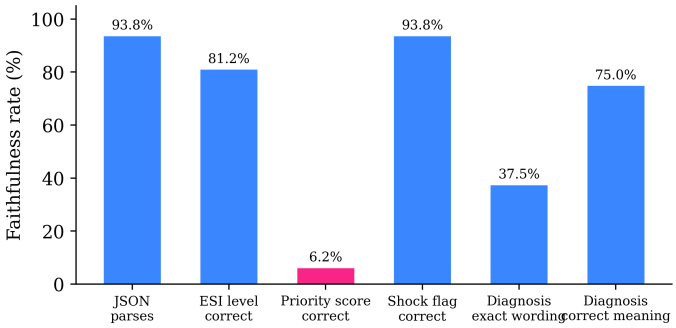


Fig. 5. LLM agent narration faithfulness by field, 16 runs. The priority score (highlighted) is the one field an earlier system revision let the LLM compute unsupervised; it is also the field the LLM restates least reliably.

VII. DISCUSSION

A. What the Numbers Actually Mean

The result in Table VI is not a defensive-programming assumption – it is a measurement that this architectural safety rule was **necessary, not merely cautious**. Had this system trusted the LLM's own JSON output for the priority score, it would have assigned the wrong queue priority to a patient in **more than 9 of every 10 runs**. Comparable agentic-orchestration architectures build in procedural guardrails against exactly this kind of failure without ever measuring how often an agent's output diverges from ground truth [14]; this is what that divergence rate looks like, measured, on a real pipeline.

The hybrid model's largest gain sits exactly where DSP-level fixes could not reach: motion artifact from speech is **orthogonal** to harmonic ambiguity, frame-rate mismatch, and history-aggregation freeze, so a learned correction absorbs a bias pattern no peak-selection heuristic can express. This also explains why bigger models did not help further: the limiting factor is **51 subjects**, not the volume of labeled cycles.

B. Positioning and Limitations

This is deliberately **not framed as an accuracy paper** competing on a single metric. Qaadan et al.'s study is a pure rPPG-accuracy benchmark with no deployed system, no clinical-safety layer, and no agent-reliability measurement; Table V states plainly that they currently win on their own metric. Real limits remain: **no real-data respiratory-rate accuracy**, since none of the three datasets provide RR ground truth; **no skin-tone-stratified accuracy**, since none ship compatible Fitzpatrick metadata; the live-camera path is **single-session**, so the dashboard's "queue" is a sequential log, not concurrent monitoring; and the clinical engine is a **custom vitals-only rule**, not a validated ESI or NEWS2 implementation [7], [8].

VIII. CONCLUSION AND FUTURE WORK

VITAL demonstrates a contactless vitals-to-triage pipeline in which a hybrid correction layer **closes 31% of the real-video accuracy gap** left by classical rPPG alone (**9.43→6.48 BPM MAE** across 71 sessions), and in which a deterministic clinical engine is kept as the **sole source of truth** for triage-critical numbers – justified by a measurement that the LLM narration layer restates a given priority score correctly only **6.2% of the time**. Three root-caused DSP bugs and one partially-fixed POS bug are reported alongside the fixes that did not work, and a real accuracy gap against the closest published benchmark is reported rather than hidden.

Future work: **closing the named feature gap** against [1] (skewness, autocorrelation, sample entropy, spectral centroid); a **structural rewrite of POS's projection step**; **skin-tone-stratified validation** once compatible metadata is available; and a **live, real-patient pilot**, deliberately out of scope here but planned.

REFERENCES

[1] S. Qaadan, R. Al Jayyousi, and O. Alkhalaileh, "A Unified Benchmarking Framework for Classical Machine Learning Based Heart Rate Estimation from RGB and NIR rPPG," *Electronics*, vol. 15, no. 1, p. 218, 2026, doi: 10.3390/electronics15010218.

[2] G. de Haan and V. Jeanne, "Robust Pulse Rate From Chrominance-Based rPPG," *IEEE Transactions on Biomedical Engineering*, vol. 60, no. 10, pp. 2878–2886, Oct. 2013, doi: 10.1109/TBME.2013.2266196.

[3] W. Wang, A. C. den Brinker, S. Stuijk, and G. de Haan, "Algorithmic Principles of Remote-PPG," *IEEE Transactions on Biomedical Engineering*, vol. 64, no. 7, pp. 1479–1491, Jul. 2017, doi: 10.1109/TBME.2016.2609282.

[4] S. Bobbia, R. Macwan, Y. Benezeth, A. Mansouri, and J. Dubois, "Unsupervised Skin Tissue Segmentation for Remote Photoplethysmography," *Pattern Recognition Letters*, 2017, doi: 10.1016/j.patrec.2017.10.017.

[5] R. Meziati Sabour *et al.*, "UBFC-Phys: A Multimodal Database for Psychophysiological Studies of Social Stress," *IEEE Transactions on Affective Computing*, 2021, doi: 10.1109/TAFFC.2021.3056960.

[6] C. Lugaresi *et al.*, "MediaPipe: A Framework for Building Perception Pipelines," *arXiv preprint arXiv:1906.08172*, 2019.

[7] N. Gilboy, T. Tanabe, D. Travers, and A. M. Rosenau, *Emergency Severity Index (ESI): A Triage Tool for Emergency Department Care, Version 4. Implementation Handbook 2012 Edition*, AHRQ Publication No. 12-0014, Agency for Healthcare Research and Quality, Rockville, MD, 2011.

[8] Royal College of Physicians, *National Early Warning Score (NEWS) 2: Standardising the Assessment of Acute-Illness Severity in the NHS. Updated Report of a Working Party*, London: RCP, 2017.

[9] P. Pirzada, A. Wilde, and D. Harris-Birtill, "Remote Photoplethysmography for Heart Rate and Blood Oxygenation Measurement: A Review," *IEEE Sensors Journal*, vol. 24, no. 15, pp. 23436–23453, Aug. 2024, doi: 10.1109/JSEN.2024.3405414.

[10] J. W. Chin *et al.*, "Clinical Validation of rPPG-Enabled Contactless Pulse Rate Monitoring Software in Cardiovascular Disease Patients," *Bioengineering*, vol. 13, no. 2, p. 246, Feb. 2026, doi: 10.3390/bioengineering13020246.

[11] B. Jonatha and F. F. Lubis, "Interpretable Machine Learning for Emergency Department Triage: A Systematic Literature Review," in *2025 15th International Conference on Information & Communication Technology and System (ICTS)*, 2025, doi: 10.1109/ICTS67612.2025.11369580.

[12] Y. Yaddaden, Y. Benahmed, M.-D. Rioux, and M. Kallel, "Machine Learning-Based Pre-Diagnosis Tools in Emergency Departments: Predicting Hospitalization, Mortality and Triage Acuity," in *2023 IEEE Third International Conference on Signal, Control and Communication (SCC)*, 2023, doi: 10.1109/SCC59637.2023.10527491.

[13] Y.-K. Liu and Y.-C. Tsai, "Explainable AI for Trustworthy Clinical Decision Support: A Case-Based Reasoning System for Nursing Assistants," in *2024 IEEE International Conference on Big Data (BigData)*, 2024, pp. 6502–6509, doi: 10.1109/BigData62323.2024.10825008.

[14] A. Warrier and Abhilash K. S., "Autonomous Agentic AI for Clinical Workflow Orchestration: Self-Managing Healthcare Operations," in *2025 6th International Conference on IoT Based Control Networks and Intelligent Systems (ICICNIS)*, 2025, pp. 1450–1457, doi: 10.1109/ICICNIS66685.2025.11315712.

[15] D. Sharma *et al.*, "Triage-Bot: An Assistive Triage Framework," in *2024 IEEE International Conference on Digital Health (ICDH)*, 2024, pp. 138–140, doi: 10.1109/ICDH62654.2024.00033.